AP CALCULUS AB

Unit 1 • Lesson 1.15 notes

# Lesson 1.15 • Connecting Limits at Infinity and Horizontal Asymptotes

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

**Learning goals**

- Evaluate limits as $x \to \pm\infty$ of rational, radical and exponential expressions.
- Use relative growth rates to compare functions.

**Vocabulary.** limit at infinity, horizontal asymptote, dominant term, growth rate.

## 1. Rational functions

Divide numerator and denominator by the highest power of $x$ in the denominator, or compare leading terms: same rules as in Precalculus.

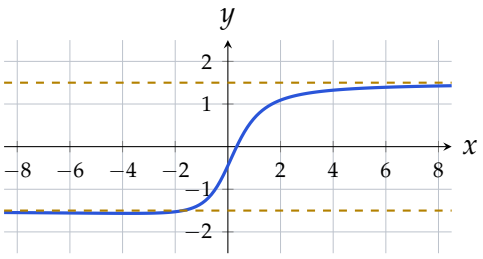
## 2. Radicals: watch the sign

$\sqrt{x^2} = |x|$, which equals $-x$ when $x < 0$.

**Example 1**

$\lim_{x\to\pm\infty} \dfrac{3x-1}{\sqrt{4x^2+5}}$.

**Solution.** For large $|x|$, $\sqrt{4x^2+5} \approx 2|x|$. As $x \to \infty$: $\frac{3x}{2x} \to \frac{3}{2}$. As $x \to -\infty$: $\frac{3x}{-2x} \to -\frac{3}{2}$. Two horizontal asymptotes: $y = \pm\frac{3}{2}$.




## 3. Growth rates

**Key idea**

As $x \to \infty$: $\ln x$ grows slower than any power $x^p$ ($p > 0$), which grows slower than any exponential $a^x$ ($a > 1$). The fastest-growing term controls the limit.

**Example 2**

**Solution.** $\lim_{x\to\infty} \dfrac{x^5}{e^x} = 0$; $\lim_{x\to\infty} \dfrac{\ln x}{\sqrt{x}} = 0$; $\lim_{x\to\infty} \dfrac{4e^x - x}{2e^x + 3} = \lim_{x\to\infty} \dfrac{4 - xe^{-x}}{2 + 3e^{-x}} = 2$.

**Common mistakes**

- Forgetting that $\sqrt{x^2} = -x$ for negative $x$, which flips the sign of the limit at $-\infty$.

- Assuming $\lim_{x\to-\infty} e^x$ is $-\infty$; it is $0$.

**Quick check**

1. $\lim_{x\to\infty} \frac{2-5x^2}{x^2+x}$ 2. $\lim_{x\to-\infty} e^x$ 3. $\lim_{x\to\infty} \frac{x^{100}}{1.01^x}$

**Answers.** 1. $-5$. 2. $0$. 3. $0$.